

Spectravideo on view

John Lettice is pleasantly surprised by the Spectravideo 328, but the MSX standard may turn into a millstone.



Waving his glass, he lurched towards me. 'Why don't you lot ever come up to Brum?' he said. 'Why don't you take a look at this wonderful machine we've got? Tell you what, I'll send you one.' Despite my protests that I didn't really want to review a Spectravideo 328, he was adamant and a few days later I found myself unpacking the beast and thinking it might just be good enough to succeed.

The Spectravideo is one of the first MSX machines, and comes in two sizes — medium and large. The 328 reviewed here has 48K ROM and 80K RAM, while the 318 has a rubber keyboard, 32K ROM and 32K RAM. They both run CP/M, though we didn't have the facilities to try this out.

MSX is intended to be a standard operating system for Z80-based micros, and is supported by a number of Japanese manufacturers; the idea is that standardisation will stimulate software development.

But the problem for the advance guard is that it'll still take time for software to be developed, and there's no guarantee that it will ever take off. Because of this there's still a great deal of scepticism about the MSX standard.

Presentation

The 328 comes in a very large box, accompanied by a subsidiary box which houses the dedicated tape recorder, and is just slightly too large for the average carrier bag. A ring-bound beginners' manual — the advanced manual has barely achieved rumour status in this country — is also included, and the review machine came with copious leaflets in Hong Kongese.

Documentation

One promo leaflet sports the howler 'includes features previously thought unspeakable', and the manual has a page headed I/O Pintouts and Memory Mays, but you can generally work out what the manual's driving at.

The manual supplied with the 328, however, was for the 318.

As far as the manual goes, it's the clearest introductory manual I've seen. It's well-illustrated and gives enough information for the beginner to write and understand simple programs by its end. It's loose-leaf, and a nice touch is the graphics screen worksheet. There are, however, a number of errors in the listings.

The pins of the expansion, cartridge, joystick and tape ports are identified, and you also get a list of ASCII codes. But, the manual is only a beginner's course, so if you want more, you'll have to wait until the *Basic Reference Guide* is on sale — our supplier couldn't say when it will be available.

Construction

The SV-328 is around the size of a standard typewriter, and unlike its little brother the 318, has full-travel typewriter keys. The detachable joystick of the 318 has been dispensed with and four cursor keys substituted in a cramped position at the top end of the numeric keypad.

The keys are too stiff for my taste, though three days of Frantic Freddy (see Software, below) loosened up the cursor keys a treat. There are five shiftable function keys above the standard qwerty layout, giving you a total of ten functions. Each function key has a function built into

the software, though they're also programmable. There are also two delete keys — the right hand one is a single character backwards delete; the left one deletes blocks forwards.

To the left of the space bar are two extra shifts which allow you access to the 52 block graphics; unfortunately, these aren't printed on the keyboard, which means something of a voyage of discovery.

There are two joystick ports down one side, and round the back are the expansion port and cassette port and cassette port. These are both non-standard edge connectors — the cassette recorder is dedicated, and built like a large brick. Its only merit is that it's dual channel, allowing you to incorporate your own speech, music or mysterious honking noises in your programs.

The bottom and the rear of the machine have large vents, through which an extremely large heat-sink can be seen, which appears to pass directly under the cartridge slot. The temperature in there gets so high I could swear I saw a heat-haze over it. The machine seems to operate happily enough, so just push down the flap and you can warm your fingers in the slot.

The machine is built to take a monitor, but incorporates two good ideas to let you use a TV. The modulator is outside the casing, so all you need do to use a monitor is get the right cable, and the TV cable itself has a switch box so you can leave it plugged in to the TV. Thoughtful details like this one are particularly heartwarming.

Operation

Both the Spectravideos run Microsoft Extended Basic, and very nice it is too. There's a built-in screen editor, a sprite

command allows easier control of user defined graphics, a double precision maths package — also built in — and banked program switching from Basic. The latter allows you to run two programs at the same time, provided you have the necessary RAMpack, but there's little information about it in the manual.

The 318 alleges 32K ROM and 32K RAM, but 16K of the latter is used for graphics, so you've only got 16K RAM. The 328 has a claimed 48K ROM and 80K RAM, and if you check memory available with PRINT FRE it gives you a figure around ... 29K? I whipped open the case with vague thoughts that I might find the missing 51K lying around in there somewhere. I then called the distributors.

Apparently there is 80K in there, but 16K of it is for graphics, and 35K is just sitting twiddling its thumbs until you access it under CP/M or Microsoft Basic on disk. The 29K available is comparable to other home micros, but Spectravideo should really put some sort of qualifier into its advertising.

The sound chip on the Spectravideos is operated by what the company calls a Music Macro Language, and responds to the basic syntax PLAY "CEG", where C, E and G are the notes of that name. You can also specify sharp or flat, and can compose on three channels. Three channels of sound come through the TV, and as the chip operates independently of the Z80 it can go its own sweet way while the on-screen action continues.

The DRAW command uses a similar syntax, eg DRAW "U50R50D50L50" — U is up, D down, R right and L left, check? But I still think calling this a Graphics Macro Language is a little overblown. You can also specify colours quite easily in DRAW commands — my only real worry was that the CIRCLE command seems to produce a very pretty ellipse.

The machine has two screen modes, one low resolution and the other 256 × 192. This seems excessively average for a new machine, but the software available does admirable things with what looks at first sight relatively low grade tools. The Spectravideo has 16 colours in all, if you include transparent, and print and background colour can be specified in the same statement. The review machine had an unpleasant screen judder whenever a white background was used, but combinations excluding white backgrounds gave good, static images.

The screen itself is 40 column, although an expansion card is available to give you the 80 columns you'd need for professional work.

Yes, the Spectravideo has a lot of software, and no, it hasn't. The reason for this is that, although the machine doesn't have a great deal of tape and cartridge software available at the moment, it does run CP/M, and therefore in theory at least the whole range of CP/M programs is available for the price of the 80 column card and disk drive.

We didn't have the necessary add-ons to check this out, and there's currently very little information about what will and will not run. The distributor says that it seems

fairly easy to reformat Kaypro software for the Spectravideo, and is currently checking this out, but it may be quite some time before full information is available.

The Spectravideo software, although thin on the ground, is impressive, especially the two cartridges we saw. Cross Force is a particularly graphic and noisy invaders type game, but it was Frantic Freddy that won hearts. You, Frantic Freddy, dodge your way through a blazing apartment building putting out fires and catching cats as they jump from the windows. It's the cats that make all the difference, and if future software is of this quality the machine's success is assured.

There are also a number of serious programs available — this is perhaps a pointer to the micro's crossover status between home and business.

Expansion

Surprisingly, most of the add-ons Spectravideo is advertising are on sale in this country. There's a motherboard expander that sits behind the machine and takes cartridges for modem (unavailable), Centronics and RS232 interfaces, 80 column screen, disk drive controller and 16K and 64K RAM-packs. The whole set-up is housed in a box which is big enough to balance a monitor on.

You can also get 5.25in floppy drives, a printer and a selection of joysticks. The graphics tablet and the Coleco Game Adaptor, which will allow you to play Coleco cartridge games, are not yet available.

The fact that the whole system is being launched at once is highly commendable, but it strikes me as a bit of a liberty to force you to spend £100+ on the motherboard just so you can get hard-copy output. Yes, you've guessed it, the SV-901 dot matrix printer has optional RS232 or Centronics. You can run a single expansion unit with the adaptor but you'll need to plug in the

motherboard if you want to run two or more of them.

The SV-328 is advertised as being expandable to 256K, feverish calculations make that the basic 80K plus three 16K and two 64K RAMpacks, taking up five of your seven slots with memory. But forget all those nightmares about grandmother boards — it's a goof in the advertising copy, and it'll only take up to 144K.

Verdict

As far as the SV-328 is concerned the talk about MSX is really something of a red herring. The machine is good enough to stand up on its own, so it might even be to Spectravideo's disadvantage if the machine's success depended on the MSX standard taking off.

A major problem is likely to be availability. There are very few machines in the country at the moment, and unless supplies increase dramatically there just won't be the user base the machine needs. At £200 the 318 is probably a bit pricey to make much of an impact, but the £300 level of the 328 could make the machine stiff competition for the BBC B, and it could well find a market as a low-end business micro.

But the real threat to the Spectravideo, and indeed to all £300 to £500 micros on the market now or about to arrive, is the new Sinclair QL.

If the Sinclair takes off as its specifications say it should, then machines like the Spectravideo will be in for a tough time.

However, from the range of peripherals, it does look like the Spectravideos have the backing they need to make a dent in the market, and logically, if the Japanese are to make an impact in home computing, they will have to resign themselves to making heavy commitments to machines like these.

As a first try, the Spectravideo is certainly a highly creditable effort, and if there's more where this came from, it's going to be an interesting year.

SPECIFICATION

Price	£302.45
Processor	Z80A
ROM	48K expandable to 96K
RAM	80K (16K of which is graphics support, and 35K accessed through CP/M)
Graphics screen	256 by 192 pixels
Keyboards	87 typewriter style keys
Storage	Tape and cartridge, single sided double density 5.25in floppy drives with 256K unformatted capacity available
Interfaces	Two joystick ports, edge-connector style expansion port
Operating system	MSX
Distributor	CK Supplies, Weston-super-Mare 418838, The Gamekeeper, 021 384 6108
Peripherals	SV-601 Super Expander £113.65 SV-903 Stereo Data Cassette £44 SV-602 Single Slot Expander £14.38 SV-902 Floppy Disk Drive £228.35 SV-901 Dot Matrix Printer £199 SV-805 RS232 Interface Cartridge £67.85 SV-803 16K RAMpack £33.35 SV-802 Centronics Interface Cartridge £67.85 SV-801 Dual disk controller £75.90

*Prices are being revised at the moment and may therefore be 3-6 per cent higher.